

(a1) Four Most Important NLP Tasks

Task	Justification
Task 1: Text classification (intent / urgency)	Needed to automatically detect the type of case (immediate danger, legal query, psychological support) from messages and speech transcripts.
Task 2: Sequence labelling (NER, slot filling)	Required to extract key entities such as names, locations, dates, and incident types for further processing and guidance.
Task 3: Retrieval / QA over documents	Enables retrieval of relevant legal information, shelters, and support resources based on detected entities and urgency.
Task 4: Multimodal fusion (combine text + speech metadata)	Integrates textual messages with speech features (tone, pauses) to improve detection of urgency and context, especially in noisy ASR transcripts.

(a2) Linguistics-aware Design Table

Task (A–F)	Main Linguistic Level	One effect on features or model design	One focus for error analysis in this domain
Text classification (intent / urgency)	Pragmatics / discourse (detect intent and urgency)	Features may include embeddings from multilingual models; model may need code-switch handling	Misclassification of urgency due to ambiguity, sarcasm, or ASR errors
Sequence labelling (NER, slot filling)	Morphology / syntax (entity boundaries and types)	Model may require token-level embeddings and CRF or transformer-based sequence labeling	Errors in entity recognition due to misspellings, ASR noise, or code-switching

(a3) Ambiguity and Pragmatics [6 Marks]

1. True

Misunderstanding domestic abuse signals can delay intervention, unlike misclassifying a product review which has low real-world risk.

2. True

Pragmatics determines whether statements imply immediate danger (e.g., indirect threats), which cannot be resolved by keywords alone.

3. False

Ambiguity is more critical here than in movie sentiment analysis because consequences involve safety and legal outcomes.

Q2 – Tokenization Strategy

1. Method Comparison (Research Design)

Method	Core Selection Principle	Strength / Weakness for Turkish/Urdu vs English
BPE	Iteratively merges most frequent symbol pairs	Strength: Good for frequent morphemes; Weakness: Rare or complex morphemes split into many subwords
WordPiece	Greedy likelihood-based subword selection	Strength: More robust for unknown words; Weakness: Can still produce long sequences for highly agglutinative words
Unigram / SentencePiece	Probabilistic subword sampling based on likelihood	Strength: Flexible segmentation; Weakness: May produce inconsistent splits in highly inflected words

2. Vocabulary vs Sequence Length Trade-off

(a) 30k vs 100k vocabulary consequences

- 30k will produce longer token sequences, 100k will produce shorter sequences
- 30k will reduce vocabulary memory footprint, 100k will increase memory/embedding size
- 30k may increase model steps per input, 100k may improve efficiency per sequence

(b) Effect on Turkish/Urdu more than English

- Morphologically rich words (Turkish/Urdu) split into more subwords with smaller vocab → longer sequences
- Agglutinative/inflected forms benefit more from larger vocab to reduce splitting and preserve semantic units

3. Evaluation Plan (Research-style criteria)

1. Average subword length per word – Measures sequence compression; important to prevent excessively long sequences for Turkish/Urdu.
2. Unknown token rate on validation corpus – Lower UNK rate shows better coverage of rare or inflected words.
3. Downstream model performance (e.g., classification/NER F1) – Directly measures practical effectiveness of tokenizer in multilingual legal tasks.

Q3

Part (A): Bag-of-Words and TF-IDF [10 Marks]

1. Vocabulary

$$V = \{fast, slow, reliable, buggy, crash, smooth\}$$
$$N = 3$$

2. BoW Counts

Feature order:

$$[fast, slow, reliable, buggy, crash, smooth]$$

d₁: “fast fast reliable app”

$$c_1 = [2, 0, 1, 0, 0, 0]$$

d₂: “slow buggy crash”

$$c_2 = [0, 1, 0, 1, 1, 0]$$

d₃: “fast reliable smooth”

$$c_3 = [1, 0, 1, 0, 0, 1]$$

3. TF Formula

$$tf(t, d) = \frac{\text{count of term } t}{\text{total terms in document}}$$

4. IDF

Document frequency:

Term	df
fast	2
slow	1
reliable	2
buggy	1
crash	1
smooth	1

IDF formula:

$$idf(t) = \log \left(\frac{N}{df(t)} \right)$$

Since N=3:

$$\log(3/2) = 0.405$$
$$\log(3) = 1.099$$

Thus:

- fast = 0.405
- slow = 1.099
- reliable = 0.405
- buggy = 1.099
- crash = 1.099
- smooth = 1.099

TF-IDF for d_1

d_1 has 3 words:

TF:

fast = $2/3$

reliable = $1/3$

Multiply by IDF:

fast: $(2/3)(0.405)=0.27$

reliable: $(1/3)(0.405)=0.135$

Final TF-IDF(d_1):

$$[0.27, 0, 0.135, 0, 0, 0]$$

Part (B): Logistic Regression [15 Marks]

Weights:

$$w = [1.0, -1.2, 0.8, -0.9, -1.1, 0.6]$$

$$b = -0.3$$

1. Prediction for d_2 **(i) Feature vector**

$$x_2 = [0, 1, 0, 1, 1, 0]$$

(ii) Compute z_2

$$\begin{aligned} z_2 &= (1.0 \times 0) + (-1.2 \times 1) + (0.8 \times 0) + (-0.9 \times 1) + (-1.1 \times 1) + (0.6 \times 0) - 0.3 \\ &= -1.2 - 0.9 - 1.1 - 0.3 \\ &= -3.5 \end{aligned}$$

(iii) Predicted probability

$$\begin{aligned} \hat{y}_2 &= \frac{1}{1 + e^{3.5}} \\ e^{3.5} &= 33.1 \\ \hat{y}_2 &= 1/34.1 = 0.029 \end{aligned}$$

≈ 0.03

Predicted class = 0

2. Cross-Entropy Loss

Since $y_2=0$:

$$\begin{aligned} L^{(2)} &= -\log(1 - 0.029) \\ &= -\log(0.971) \\ &= 0.029 \end{aligned}$$

Loss ≈ 0.03

3. Confusion Matrix

True: [1, 0, 1, 0]

Pred: [1, 1, 0, 0]

TP=1

FP=1

TN=1

FN=1

Precision = $1/(1+1)=0.5$

Recall = $1/(1+1)=0.5$

F1 = 0.5

Final Key Results

- TF-IDF(d_1) = [0.27, 0, 0.135, 0, 0, 0]
- $z_2 = -3.5$
- $\hat{y}_2 \approx 0.03$
- Loss ≈ 0.03
- Precision = Recall = F1 = 0.5